ST6

Air separation with membranes

Oxygen selective prism membranes

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1 Introduction

The goal of this experiment is to study the separation of oxygen and nitrogen from air using membrane technology. Membrane gas separation is a common technique in industrial processes, including nitrogen generation and oxygen enrichment, by taking advantage of differences in gas permeability through selective membranes. This experiment will examine how factors such as membrane selectivity, pressure ratio, and stage cut influence the efficiency of the separation process.

2 Theory

2.1 Backround theory

A membrane acts as a selective barrier between two phases—in the case of air separation, these phases are both gas. The separation process occurs because one component of the gas mixture passes through the membrane more readily and quickly than the others.

Both porous and dense membranes can be used as selective barriers, with their effectiveness depending largely on pore size. To achieve separation, pore sizes must be smaller than 0.1 m. However, non-porous polymeric membranes are the most widely used for gas separation processes today.

Three key properties influence membrane separation: selectivity, diffusion coefficient, and sorption. While the permeability describes how easily a gas component moves through the membrane, the selectivity indicates the membrane's ability to differentiate between two components. The diffusion coefficient describes how readily gas molecules move through the membrane material, affecting the rate of diffusion. Sorption (or solubility coefficient) indicates the amount of gas that can dissolve within the membrane, which depends on the gases' condensation properties and impacts the overall gas transport through the membrane.

In a membrane-based gas separation process, three streams are present: the feed gas, the permeate (the gas that has passed through the membrane), and the retentate (the gas that remains behind).

2.2 Equations

The overall material balance for the separation is shown in Eq. (2.1).

$$L_f = L_o + V_p \tag{2.1}$$

Where L_f, L_o and V_p are respectively the feed flow rate, retentate flow rate and the permeate flow rate.

The cut rate θ is a parameter of economic importance in commercial membrane processes and is shown in Eq. (2.2). It directly influences the purity and yield of the product.

$$\theta = \frac{V_p}{L_f} \tag{2.2}$$

Recovery R, shown in Eq. (2.3), is a parameter that describes how much of a specific gas component A is retained in the product stream relative to the amount in the original feed stream. This indicates how effectively the membrane process separates and holds back the component. A higher R value means that a larger portion of the component is retained in the retentate (does not go through the membrane).

$$R_A = \frac{X_{o,A} L_o}{X_{f,A} L_f} \cdot 100\% \tag{2.3}$$

Here $X_{o,A}$, L_o, $X_{f,A}$ and L_f are the mole fraction (X) and flow rate (L) of the retentate (o) and feed stream (f). By inserting the corresponding mole fraction and flow rate Eq. (2.3) can be used to calculate the recovery of O_2 and N_2.

How to calculate the mole fraction of oxygen in the feed stream concentration in the feed (X_{f,O_2}) is shown in Eq. (2.4).

$$X_{f,O_2} = \frac{L_o \cdot X_{o,O_2} + V_p \cdot X_{o,O_2}}{L_f} \tag{2.4}$$

Here X_{o,O_2} is the mole fraction of oxygen in the retentate stream andV_p the flow rate of the permeate stream.

The pressure ratio varphi is important when it comes to membrane separation because a flow of component i can only occur if the partial pressure of component i on the feed side is greater than the partial pressure on the permeate side. [1]

$$\varphi = \frac{p_o}{p_l} \tag{2.5}$$

The separation is driven by a pressure differential across the membrane. Compressed air is fed into one side of the membrane module. The higher partial pressure on the feed side pushes oxygen molecules through the membrane faster than nitrogen. The oxygen-enriched stream emerges on the permeate side, while nitrogen and other gases remain on the feed side and exit as a residual stream.

2.3 Permeate- & Repitate- pressure

The permeate should be oxygen-enriched. Since the membrane is selective for oxygen, oxygen molecules diffuse through the membrane faster than nitrogen or other gases. Therefore, the permeate stream will have a higher concentration of oxygen compared to the feed air. However, the oxygen concentration typically ranges between 25-40%, not pure oxygen. The retentate should be nitrogen-rich. As oxygen selectively permeates through the membrane, nitrogen, along with other less-permeable gases, remains in the retentate stream. This means the retentate contains a higher percentage of nitrogen than the original feed air. [2]

2.4 Counter-current flow

In counter-current flow, the feed gas and the permeate gas stream move in opposite directions through the membrane. This configuration provides a higher separation efficiency because it maintains a strong concentration gradient across the membrane for a longer portion of the module. As the feed gas continuously encounters fresher permeate, the driving force for oxygen permeation remains steady, resulting in better separation and higher oxygen enrichment in the permeate stream. This flow setup is ideal for applications requiring high-purity oxygen or nitrogen and maximized separation performance. [3]

2.5 Co-current flow

In co-current flow, both the feed gas and permeate stream move in the same direction through the membrane. Initially, the concentration gradient is large, creating a strong driving force for permeation, but this gradient diminishes rapidly as the gas flows along the membrane, reducing the separation efficiency. As a result, co-current flow typically leads to lower oxygen purity in the permeate and less efficient separation overall, though it can be faster. This flow arrangement is useful in simpler designs where speed is more critical than maximum gas purity. [3]

2.6 Cross-flow

In cross-flow operation, the feed gas flows tangentially across the membrane's surface while the permeate gas moves perpendicular through the membrane. This configuration maintains a continuous concentration gradient, though not as efficiently as in counter-current systems. Cross-flow ensures better membrane surface utilization, preventing fouling and maintaining stable performance over time. The result is moderate separation efficiency and oxygen purity, making it well-suited for industrial applications where membrane durability and steady, continuous gas separation are important. [3]

3 Experimental

Four membrane modules of polysulfone, with an oxygen-selective membrane will be used. These modules can be arranged in different configurations such as series or parallel. Flow meters and oxygen analyzers will measure gas streams at different points in the system. . A description of what you will do in the experiment, operating procedure and what do you expect for every configuration.

- Close the main feed valve and open the gas outled, set the pressure at to approximately 5 bar
- Check for leaks
- Set the flow to be testet on the flow controller and check value of the mass flow controller in retentate stream
- Slowly open the main feed valve in the set-up
- When stable state; start the sampling, and stop it when the run is finished
- After the last run close the main valve, and check the feed pressure close to 1 bar.
- Repeat these steps and record data at different flow rates and membrane configurations (series/parallel).

References

[1] Richard W. Baker. *Membrane Technology and Applications.* Wiley, 2004.

[2] Ernest J. Henley D. Keith Roper Seader, J. D. *Separation Process Principles.* Wiley, 2016. s.438-431.

[3] M Da Conceicao. Gas separation membrane module modeling: A comprehensive review. *MDPI*, 2023.

A Example calculations

In order to do the example calculations, the following data was given:

- Retentate flow rate: 21.3 L/s
- Permeate flow rate: 4.1 L/s
- O_2 concentration in retentate: 16.0%
- O_2 concentration in permeate: 40.7%
- Feed pressure: 4.7 bar
- Permeate pressure: 0.9 bar
- Retentate pressure: 4.6 bar

A.1 Feed Flow Rate

The total feed flow rate, L_f, can be determined using the overall material balance

$$L_f = L_o + V_p \tag{A.1}$$

Where L_f is the feed flow rate, L_o is the retentate flow rate, and V_p is the permeate flow rate. When inserting the given values into the equation A.1, we get the following

$$L_f = 21.3 + 4.1 = 25.4\,\mathrm{L/s} \tag{A.2}$$

A.2 Cut Rate

The cut rate, θ, is given by the following equation

$$\theta = \frac{V_p}{L_f} \tag{A.3}$$

When inserting the given values, the following value is calculated

$$\theta = \frac{4.1}{25.4} = 0.161 \tag{A.4}$$

A.3 Pressure Ratio

The pressure ratio, φ, is the ratio of the feed pressure,P_o, to the permeate pressure, P_l

$$\varphi = \frac{P_{\text{feed}}}{P_{\text{permeate}}} \tag{A.5}$$

Substituting the given values gives

$$\varphi = \frac{4.7}{0.9} = 5.22 \tag{A.6}$$

A.4 Recovery of O_2

The recovery of oxygen, R_{O_2}, is calculated using the following equation

$$R_{O_2} = \frac{x_{o,O_2} \cdot L_o}{x_{f,O_2} \cdot L_f} \cdot 100 \tag{A.7}$$

where x_{o,O_2} is the mole fraction of O_2 in the retentate, and x_{f,O_2} is the mole fraction of O_2 in the feed. By substituting the given values into the equation A.7, the following is calculated

$$R_{O_2} = \frac{16.0 \cdot 21.3}{40.7 \cdot 25.4} \cdot 100 = 33.2\% \tag{A.8}$$

A.5 Recovery of N_2

For nitrogen, it is assumed that N_2 is the balance of the gas composition, i.e., 100% - O_2%. To calculate R_{N_2}, the same equation used for calculating R_{O_2} can be used, A.7

$$R_{N_2} = \frac{0.84 \cdot L_o}{0.8001 \cdot L_f} \cdot 100 \tag{A.9}$$

By substituting the values the following calculations can be done

$$R_{N_2} = \frac{0.84 \cdot 21.3}{0.8001 \cdot 25.4} \cdot 100 \approx 88.04\% \tag{A.10}$$

B HSE

The largest risk is equiptment breakage caused by overpressure. If it occurs in the system it can cause leakage or damage both on the system, and in worst case projecting glass spliters. Methods to mitigate are to always monitor pressure gauges and ensure proper valve operation. And double-check connections and ensure proper flow before starting the experiment.